WORKED SOLUTION

STEP II 1995, Question 6

Pure • Roots of Unity

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q6

The key idea

The seventh roots of unity are $1, \alpha, \alpha^2, \ldots, \alpha^6$, and they add up to zero. Split the six non-real ones into two groups of three, $\{\alpha, \alpha^2, \alpha^4\}$ and $\{\alpha^3, \alpha^5, \alpha^6\}$. The second group is just the complex conjugates of the first, so the two group sums are conjugates. Their sum and product are therefore real, and the first part turns them into the coefficients of the quadratic. Taking real and imaginary parts of the root then gives the trigonometric sums.

Solution

Roots and coefficients

If u and v are the roots of $z^2 + az + b = 0$, then the monic quadratic $z^2 + az + b$ has the factors $z - u$ and $z - v$, so

$$z^2 + az + b = (z - u)(z - v) = z^2 - (u + v)z + uv$$

for all z. Comparing coefficients of z and the constant terms gives $a = -u - v$ and $b = uv$, as required.

The roots of unity

A word of warning first. The paper says $z^6 - 1 = 0$, but this is a misprint for $z^7 - 1 = 0$. Indeed, $\alpha^6 = \cos(12\pi/7) + \mathrm{i}\sin(12\pi/7) \neq 1$, while the rest of the question is plainly about sevenths. We work with $z^7 - 1 = 0$.

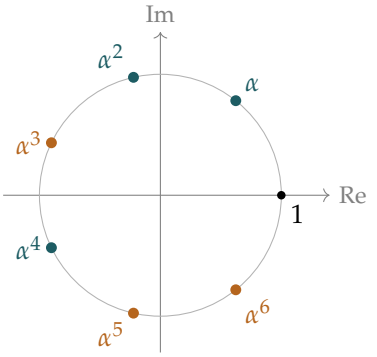
By de Moivre's theorem,

$$\alpha^7 = \cos 2\pi + \mathrm{i}\sin 2\pi = 1,$$

so α is a root of $z^7 - 1 = 0$. More generally, $\alpha^k = \cos(2k\pi/7) + \mathrm{i}\sin(2k\pi/7)$ and $(\alpha^k)^7 = (\alpha^7)^k = 1$. The seven numbers

$$1,\ \alpha,\ \alpha^2,\ \alpha^3,\ \alpha^4,\ \alpha^5,\ \alpha^6$$

are distinct, since their arguments $2k\pi/7$ for $k = 0, \ldots, 6$ are distinct in $[0, 2\pi)$. A polynomial of degree 7 has at most 7 roots, so these are all the roots of $z^7 - 1 = 0$. They sit at the vertices of a regular heptagon on the unit circle.



Two facts do all the work. First, the roots sum to zero: $z^7 - 1 = (z-1)(z^6 + z^5 + \cdots + z + 1)$, and $\alpha \neq 1$, so

$$1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = 0. \qquad (*)$$

(Equivalently, the coefficient of z^6 in $z^7 - 1$ is zero.) Second, since $|\alpha| = 1$ we have $\overline{\alpha^k} = \alpha^{-k} = \alpha^{7-k}$. So $\alpha^6 = \overline{\alpha}$, $\alpha^5 = \overline{\alpha^2}$ and $\alpha^3 = \overline{\alpha^4}$.

Finding A and B

Let

$$p = \alpha + \alpha^2 + \alpha^4, \qquad q = \alpha^3 + \alpha^5 + \alpha^6.$$

By the conjugate pairs above, $q = \overline{p}$. That is the natural guess for the other root: a quadratic with real coefficients and a non-real root p must also have $\overline{p}$ as a root.

Ask yourself How do we know p isn't real? If it were, the "other root" needn't be $\overline{p}$.

The imaginary part of p is $\sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7}$, which we show at the end is positive. So p is not real and the roots are p and $\overline{p} = q$. (In any case, once we have found real A and B with p and q as roots, the quadratic is determined.)

By $(*)$,

$$p + q = \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = -1.$$

For the product, expand and reduce the powers using $\alpha^7 = 1$:

$$\begin{aligned} pq &= (\alpha + \alpha^2 + \alpha^4)(\alpha^3 + \alpha^5 + \alpha^6) \\ &= \alpha^4 + \alpha^6 + \alpha^7 + \alpha^5 + \alpha^7 + \alpha^8 + \alpha^7 + \alpha^9 + \alpha^{10} \\ &= \alpha^4 + \alpha^6 + 1 + \alpha^5 + 1 + \alpha + 1 + \alpha^2 + \alpha^3 \\ &= 3 + (\alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6) = 3 - 1 = 2. \end{aligned}$$

Note that each of $\alpha, \ldots, \alpha^6$ appears exactly once in the nine terms, besides three 1s. By the first part, $A = -(p+q)$ and $B = pq$, so

$$A = 1, \qquad B = 2,$$

and p and q are the roots of $z^2 + z + 2 = 0$, namely

$$z = \frac{-1 \pm \mathrm{i}\sqrt{7}}{2}.$$

The trigonometric sums

Taking real and imaginary parts of $p = \alpha + \alpha^2 + \alpha^4$, with $\alpha^k = \cos(2k\pi/7) + \mathrm{i}\sin(2k\pi/7)$, gives

$$\operatorname{Re} p = \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7},$$
$$\operatorname{Im} p = \sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7}.$$

Both roots of $z^2 + z + 2 = 0$ have real part $-\frac{1}{2}$, so

$$\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7} = -\frac{1}{2},$$

as required. (Alternatively, $p + \overline{p} = -1$ gives $2\operatorname{Re} p = -1$ directly.)

The imaginary part is $\pm\frac{\sqrt{7}}{2}$, and we must decide which.

> **Ask yourself** Two of the three sines are positive and one is negative. How can we be sure which wins?

Compare the negative term with one of the positive ones. Since $\frac{8\pi}{7} = \pi + \frac{\pi}{7}$, we have $\sin\frac{8\pi}{7} = -\sin\frac{\pi}{7}$. Also, $\sin\frac{4\pi}{7} = \sin\left(\pi - \frac{4\pi}{7}\right) = \sin\frac{3\pi}{7}$. Sine is strictly increasing on $[0, \frac{\pi}{2}]$ and $0 < \frac{\pi}{7} < \frac{3\pi}{7} < \frac{\pi}{2}$, so $\sin\frac{3\pi}{7} > \sin\frac{\pi}{7}$. Hence

$$\sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7} = \underbrace{\sin\frac{2\pi}{7}}_{>0} + \underbrace{\left(\sin\frac{3\pi}{7} - \sin\frac{\pi}{7}\right)}_{>0} > 0,$$

and therefore

$$\sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7} = \frac{\sqrt{7}}{2}.$$

This also confirms that $p = \frac{-1+\mathrm{i}\sqrt{7}}{2}$ is not real, as we used above.

Checks

On a calculator (in radians), $\cos\frac{2\pi}{7} \approx 0.6235$, $\cos\frac{4\pi}{7} \approx -0.2225$ and $\cos\frac{8\pi}{7} \approx -0.9010$, which add to -0.5000. The sines are about 0.7818, 0.9749 and -0.4339, which add to 1.3229. That matches $\frac{\sqrt{7}}{2} \approx 1.3229$. You can also check that $p^2 + p + 2 = 0$ directly: with $p = \frac{-1+\mathrm{i}\sqrt{7}}{2}$, $p^2 = \frac{-6-2\mathrm{i}\sqrt{7}}{4}$, and $p^2 + p + 2 = 0$ follows.

Common slips

- Using $z^6 - 1 = 0$ as printed. Then α isn't a root, and the sum-to-zero fact $(*)$ fails.
- Forgetting to reduce powers with $\alpha^7 = 1$ when expanding pq, or miscounting the three terms equal to 1.
- Asserting that the sine sum is positive "because two terms are positive". You need a comparison such as $\sin\frac{3\pi}{7} > \sin\frac{\pi}{7}$.

Beyond STEP

The split $\{1,2,4\}$ versus $\{3,5,6\}$ is not arbitrary. The numbers $1,2,4$ are the non-zero squares modulo 7, and $2p+1 = \mathrm{i}\sqrt{7}$ is a *Gauss sum*. Gauss showed that for a prime ℓ of the form $4k+3$, the sum of α^r over the quadratic residues r is $\frac{-1+\mathrm{i}\sqrt{\ell}}{2}$. This is the first step of the theory of which regular polygons can be constructed with ruler and compass (Galois theory) and of quadratic reciprocity. Davenport's *The Higher Arithmetic* covers quadratic residues, and Needham's *Visual Complex Analysis* gives a geometric view of roots of unity.

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